STA2002: Probability and Statistics II
Likelihood Ratio Tests

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Introduction

- In this lecture we consider a general test-construction method, which is the so-called **likelihood ratio tests**, that is applicable when either or both of the null and alternative hypotheses H_0 and H_1 are composite.

  Remark. The statistics constructed for ANOVA, which have been introduced previously, are actually likelihood ratio statistics.

- **Suggested reading**: Chapter 8.7 of the textbook.

Introduction

- The hypotheses will be stated in general as

$$H_0 : \theta \in \omega \text{ v.s. } H_1 : \theta \in \omega',$$

where ω is a subset of Ω and ω' is the **complement** of ω with respect to Ω.

- Ω denotes the **parameter space** which is the set of all possible values of the parameter θ.
- The test will be constructed with the use of a ratio of likelihood functions which would be maximized in ω and Ω, respectively.

Likelihood Ratio Statistic

Definition 8.7-1

The **likelihood ratio** is the quotient

$$\Lambda := \frac{L(\hat{\omega}; X_1, \ldots, X_n)}{L(\hat{\Omega}; X_1, \ldots, X_n)}$$

where $L(\hat{\omega})$ is the maximum of the likelihood function with respect to θ when $\theta \in \omega$ and $L(\hat{\Omega})$ is the maximum of the likelihood function with respect to θ when $\theta \in \Omega$.

- Note that, Λ is a function of a random sample $X_1, \ldots, X_n$, thus Λ is random.
- When observe realized value $x_1, \ldots, x_n$, the corresponding realized value of Λ is denoted as λ.

Intuition and Observation

- Because Λ is the quotient of nonnegative functions, thus always, $\Lambda \geq 0$.
- In addition, since $\omega \subset \Omega$, it follows that

$$L(\hat{\omega}) \leq L(\hat{\Omega}),$$

  hence $\Lambda \leq 1$.
- Thus, always, $0 \leq \Lambda \leq 1$.
- **Observation**: A small value of the ratio Λ would lead to the rejection of H_0. In contrast, a value of the ratio Λ that is close to 1 would support the null hypothesis H_0.

Form of the Critical Region

Definition 8.7-2

To test

$$H_0 : \theta \in \omega \text{ v.s. } H_1 : \theta \in \omega',$$

the **critical region** for the **likelihood ratio test** is the set of points in the sample space for which

$C := \{(x_0, x_1, \ldots) \mid \Lambda_0 \leq k\}$

$$\Lambda = \frac{L(\hat{\omega})}{L(\hat{\Omega})} \leq k$$

where $0 < k < 1$ and k is selected so that the test has a desired significance level α.

An Example

Example 8.7-1

Assume that the weight X in ounces of a “10-pound" bag of sugar is $N(\mu, 5)$. We would test the hypothesis $H_0 : \mu = 162$ against the alternative hypothesis $H_1 : \mu \neq 162$. The parameter space is $\Omega = \mathbb{R}$ and $\omega = \{162\}$.

Task:
To derive the form of critical region by using the likelihood ratio statistic.

An Example (Cont'd)

- Solution. Prepare the likelihood function $L(\mu)$, and also the expression of

$$\begin{aligned}\Lambda &= \frac{L(\hat{\omega}; X_1, \ldots, X_n)}{L(\hat{\Omega}; X_1, \ldots, X_n)} \\ &= \exp\left\{-\frac{n}{10}(\overline{X} - 162)^2\right\}.\end{aligned}$$

- If replace $X_1, \ldots, X_n$ by the realized numbers $x_1, \ldots, x_n$, then we get λ,

$$\begin{aligned}\lambda &= \frac{L(\hat{\omega}; x_1, \ldots, x_n)}{L(\hat{\Omega}; x_1, \ldots, x_n)} \\ &= \exp\left\{-\frac{n}{10}(\overline{x} - 162)^2\right\}.\end{aligned}$$

An Example (Cont'd)

$$\lambda = \frac{(10\pi)^{-n/2}\exp\left[-\left(\frac{1}{10}\right)\sum_{i=1}^{n}(x_i-162)^2\right]}{(10\pi)^{-n/2}\exp\left[-\left(\frac{1}{10}\right)\sum_{i=1}^{n}(x_i-\bar{x})^2\right]}$$

$$= \frac{\exp\left[-\left(\frac{1}{10}\right)\sum_{i=1}^{n}(x_i-\bar{x})^2-\left(\frac{n}{10}\right)(\bar{x}-162)^2\right]}{\exp\left[-\left(\frac{1}{10}\right)\sum_{i=1}^{n}(x_i-\bar{x})^2\right]}$$

$$= \exp\left[-\frac{n}{10}(\bar{x}-162)^2\right].$$

$\sum(x_i-162)^2$

$= \sum(x_i-\bar{x}+\bar{x}-162)^2$

$= \sum(x_i-\bar{x})^2 + n\cdot(\bar{x}-162)^2$

An Example (Cont'd)

- On one hand, a value of $\overline{x}$ close to 162 would tend to support H_0, and in that case λ is close to 1.
- On the other hand, a value of $\overline{x}$ that differs from 162 by too much would tend to support H_1.
- The form of a critical region C for a likelihood ratio test is given by

$$C := \{(X_1, X_2, \ldots, X_n) \mid \Lambda \leq k\},$$

where k **is selected so that the significance level of the test is** α, that is,

$$P(\Lambda \leq k;\ H_0) = \alpha.$$

An Example (Cont'd)

- Try to derive further and show that, $\alpha = P_{H_0}(C) = P_{H_0}(\Lambda_n \le k)$

$$P(\Lambda \le k;\ H_0) = \alpha$$

under H_0, $\mu = 162$. $\bar{X} \sim N(162, \frac{\sigma^2}{n})$

would yield the following critical region form,

$$C = \left\{ (X_1, X_2, \ldots, X_n) :\ |\overline{X} - 162| \ge z_{\alpha/2} \frac{\sqrt{5}}{\sqrt{n}} \right\}.$$

- **Remark**. The inequality $\Lambda \le k$ can often be expressed in terms of a statistic whose distribution is known. For example, here, the statistic is $\overline{X}$ which follows a normal distribution.

Tests of Variance Again

- Next, we try to construct a test about the variance of a normal population.
- Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$, where μ **and** σ^2 **are unknown**.
- We would test

$$H_0 : \ \sigma^2 = \sigma_0^2 \ \text{ against } \ H_1 : \ \sigma^2 \neq \sigma_0^2.$$

- Question: How to use likelihood ratio statistic to construct a critical region?

To Construct a Critical Region

- We first specify the set of ω and Ω as follows,

$$\omega = \left\{(\mu, \sigma^2) : \ \mu \in \mathbb{R}, \ \sigma^2 = \sigma_0^2\right\},$$

$$\Omega = \left\{(\mu, \sigma^2) : \ \mu \in \mathbb{R}, \ \sigma^2 > 0\right\}.$$

- Then prepare the likelihood ratio, (try the deduction... we would show in class)

$$\Lambda = \left(\frac{W}{n}\right)^{n/2} \exp\left\{-\frac{W}{2} + \frac{n}{2}\right\} \leq k$$

with $W = \sum_{i=1}^{n}(X_i - \overline{X})^2/\sigma_0^2$.

To Construct a Critical Region

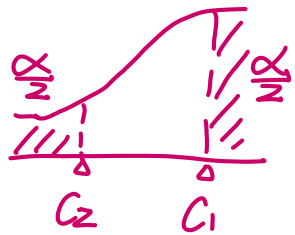
- The inequality $\Lambda \leq k$ would yield,

$$W \leq c_1 \quad \text{or} \quad W \geq c_2,$$

where c_1 and c_2 are appropriate constants so as to achieve the desired significance level α.

- Note that, under $H_0 : \sigma^2 = \sigma_0^2$, $W = \sum_{i=1}^{n}(X_i - \overline{X})^2/\sigma_0^2$ follows a chi-square distribution with d.f. $= n - 1$.
- We can set

$$c_1 = \chi^2_{1-\alpha/2}(n-1),$$



and

$$c_2 = \chi^2_{\alpha/2}(n-1).$$

Tests of Mean Again

- Next, we try to construct a test about the mean of a normal population.
- Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$, where μ **and** σ^2 **are unknown**.
- We would test

$$H_0 : \ \mu = \mu_0 \ \text{ against } \ H_1 : \ \mu \neq \mu_0.$$

- Question: How to use likelihood ratio statistic to construct a critical region?

To Construct a Critical Region

- We first specify the set of ω and Ω as follows,

$$\omega = \{(\mu, \sigma^2) : \ \mu = \mu_0, \ \sigma^2 > 0\},$$

$$\Omega = \{(\mu, \sigma^2) : \ \mu \in \mathbb{R}, \ \sigma^2 > 0\}.$$

- Then prepare the likelihood ratio, (try the deduction... we would show in class)

$$\Lambda = \left[\frac{1}{1 + \frac{n(\overline{X} - \mu_0)^2}{\sum_{i=1}^{n}(X_i - \overline{X})^2}} \right]^{n/2}.$$

To Construct a Critical Region

- The inequality $\Lambda \leq k$ would yield,

$$\frac{n(\overline{X}-\mu_0)^2}{\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\overline{X})^2} \geq (n-1)(k^{-2/n}-1).$$

- Note that, under $H_0 : \mu = \mu_0$, $\sqrt{n}(\overline{X}-\mu_0)/\sigma$ is $N(0,1)$ and $\sum_{i=1}^{n}(X_i-\overline{X})^2/\sigma^2$ has an independent chi-square distribution $\chi^2(n-1)$.
- Thus, we can construct the following statistic,

$$T = \frac{\sqrt{n}(\overline{X}-\mu_0)/\sigma}{\sqrt{\sum_{i=1}^{n}(X_i-\overline{X})^2/[\sigma^2(n-1)]}} = \frac{\overline{X}-\mu_0}{S/\sqrt{n}}.$$

To Construct a Critical Region

- Under H_0, T follows a t distribution with d.f. $= n - 1$.
- In accordance with the likelihood ratio test criterion, H_0 is rejected if the observed statistic,

$$T^2 \geq (n-1)(k^{-2/n} - 1).$$

- The critical region at significance level α is of the following form,

$$C = \{(X_1, \ldots, X_n) : \ |T| \geq t_{\alpha/2}(n-1)\}.$$

Practice Example

LR Test for the Exponential Distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from the exponential distribution with pdf,

$$f(x;\theta) = \theta^{-1} \exp\{-x/\theta\}, \quad x, \theta > 0.$$

Task:

To find a likelihood ratio test critical region of size α for testing the hypotheses

$$H_0 : \theta = \theta_0,$$

against

$$H_1 : \theta \neq \theta_0.$$